

Image Inpainting Through GNN

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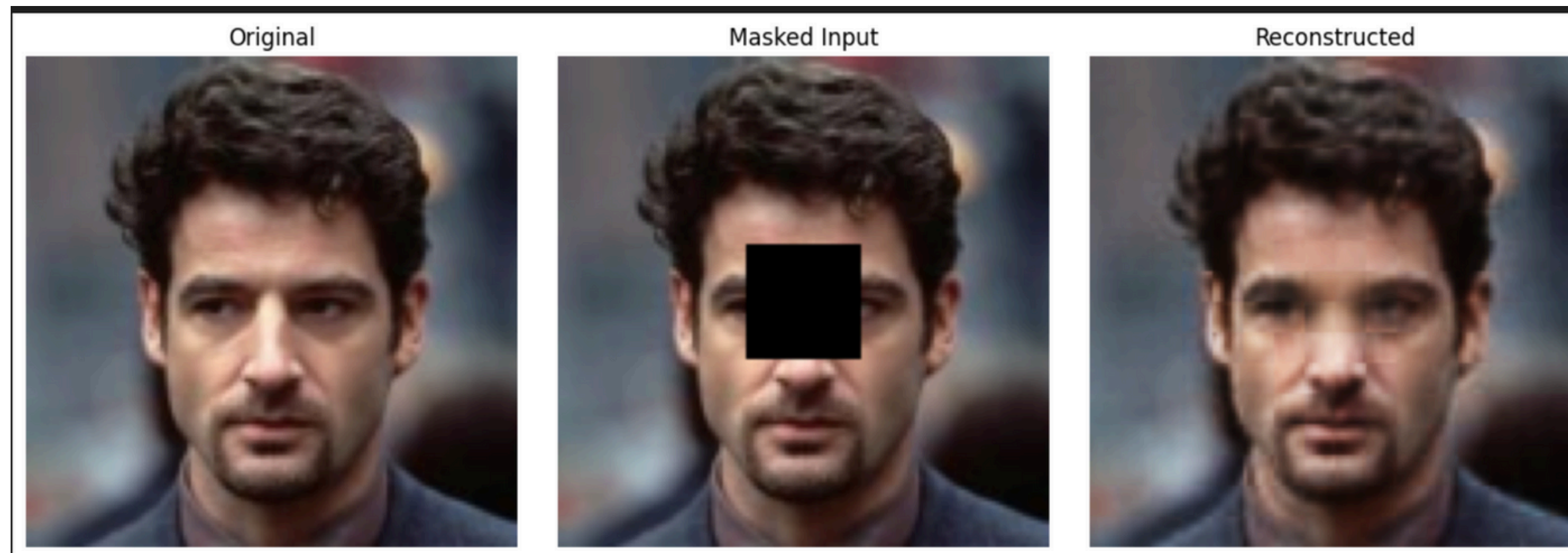
Date: March 30 2025

01

Introduction / Motivation

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- ✓ Image restoration is the process of recovering or reconstructing a damaged, missing, or corrupted part of an image to make it look as close as possible to the original version.



Introduction / Motivation



Human Face Image Restoration

Recovering damaged or low-resolution facial images is important in areas such as identity verification, security surveillance and historical photo recovery.



Medical Image Restoration

Restoring missing or unclear regions can lead to better diagnosis and treatment planning.



Satellite Image Restoration

Clearer images can improve environmental monitoring and support faster, more accurate disaster response.

02

Dataset / Models

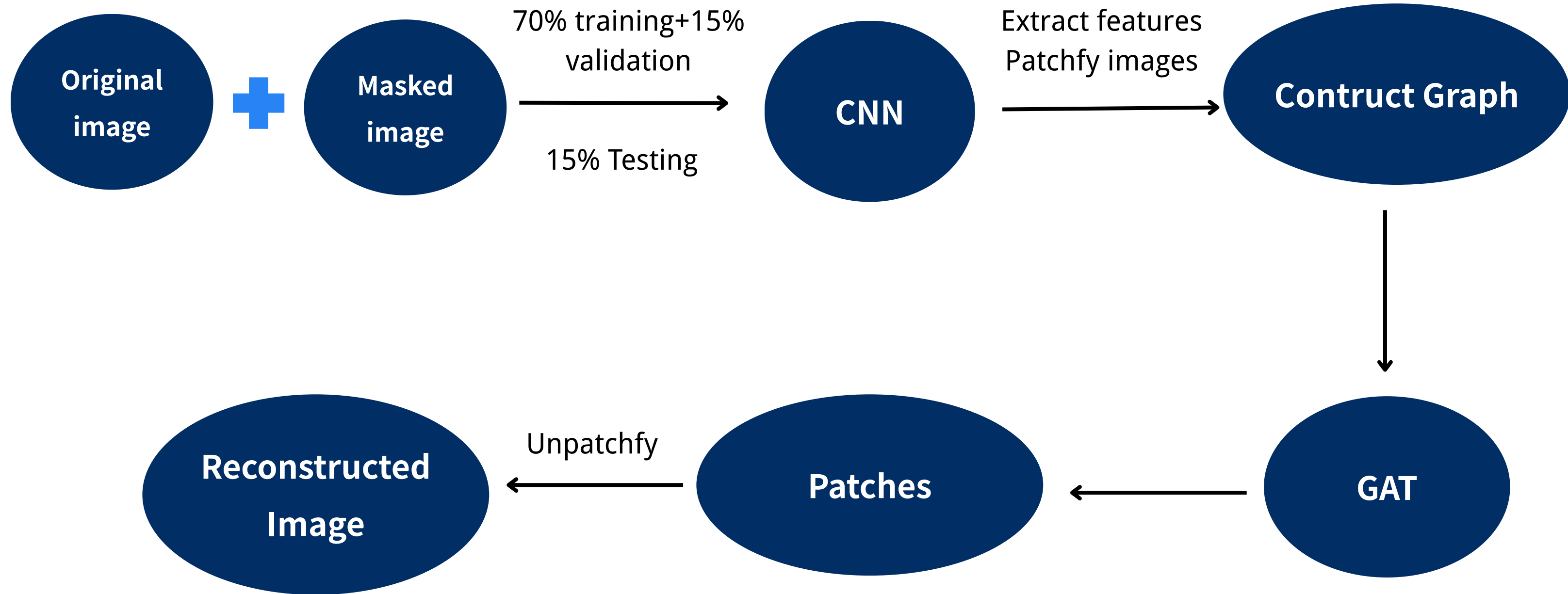
Dataset

CelebA (CelebFaces Attributes) dataset

It is a large-scale face image dataset widely used in computer vision tasks such as face recognition, face attribute prediction, image generation, and image inpainting.



Model Architecture



03

Result

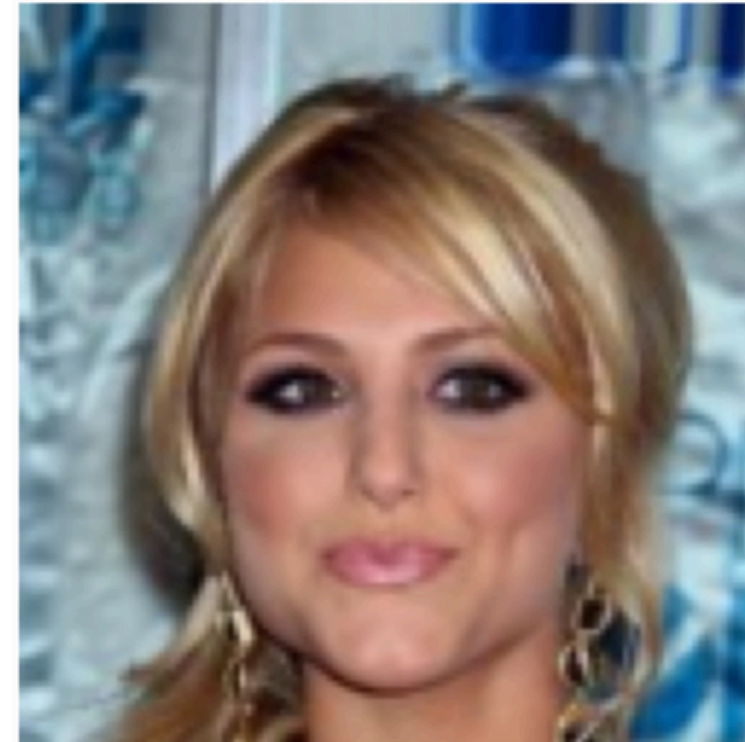
- **Results of Three Models:**

Model	PSNR	SSIM
CNN+GAT	30.39	0.92
Unet Convolutional AutoEncoder	34.48	0.98

Result

CNN+GAT

Original



Masked Input



Reconstructed



Unet
AutoEncoder

Original



Masked Input



Reconstructed



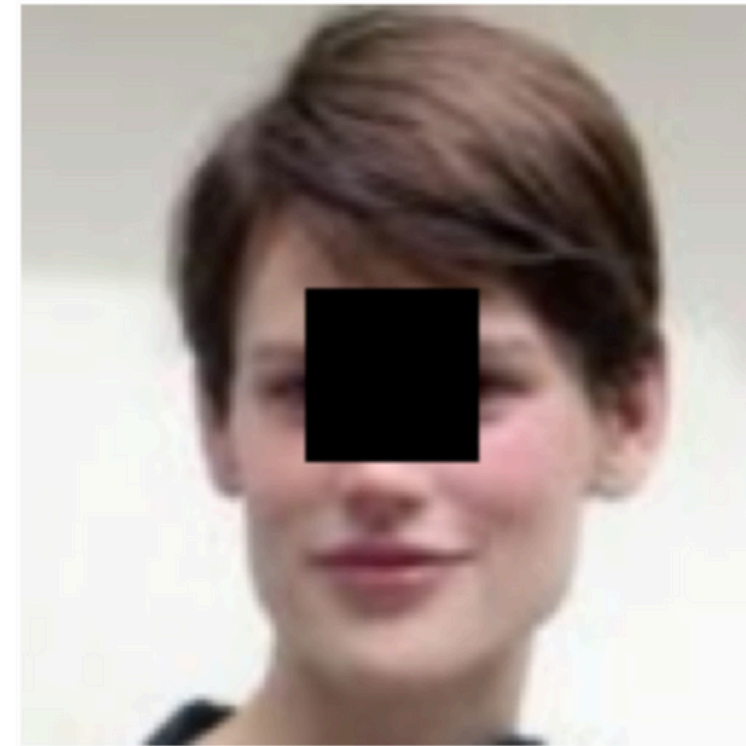
Result

CNN+GAT

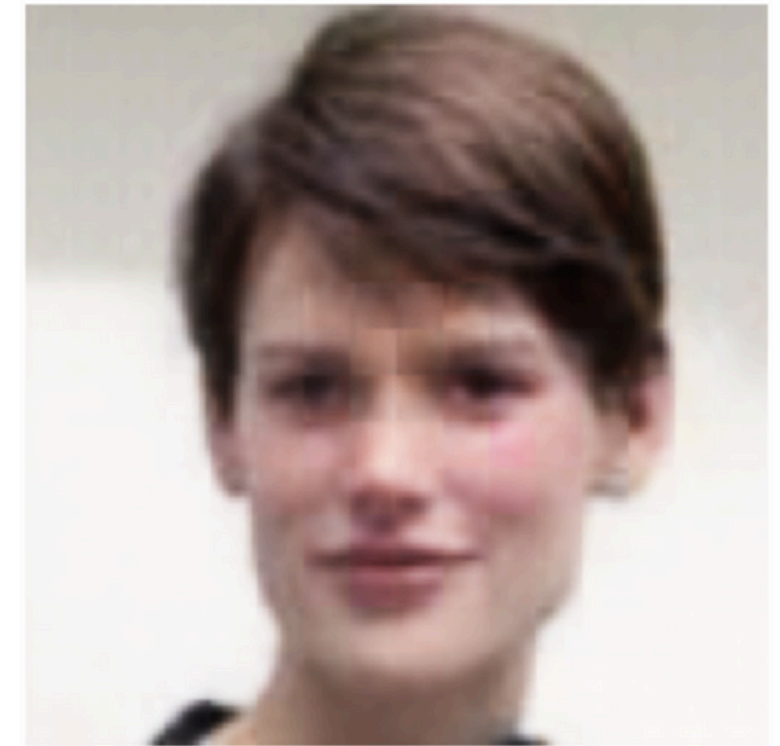
Original



Masked Input



Reconstructed

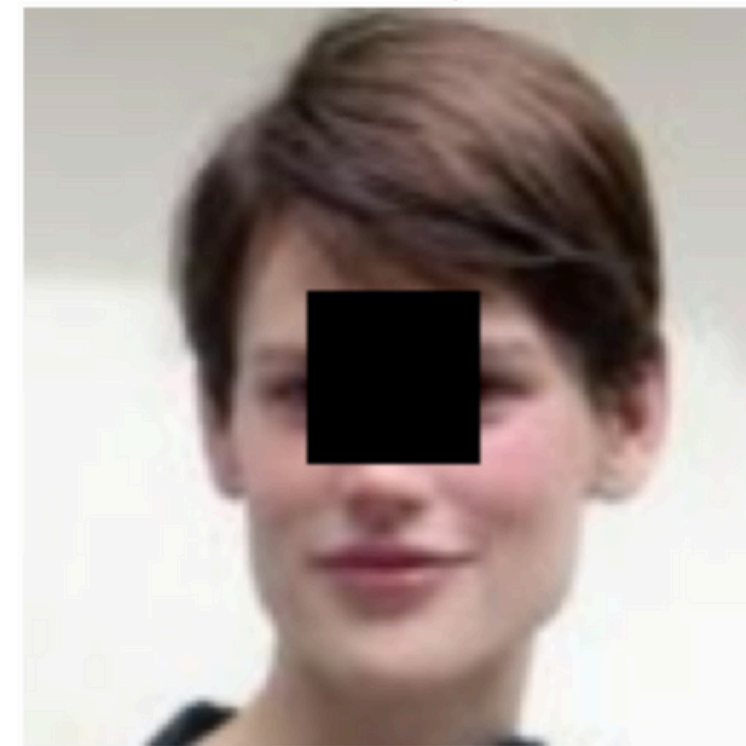


Unet
AutoEncoder

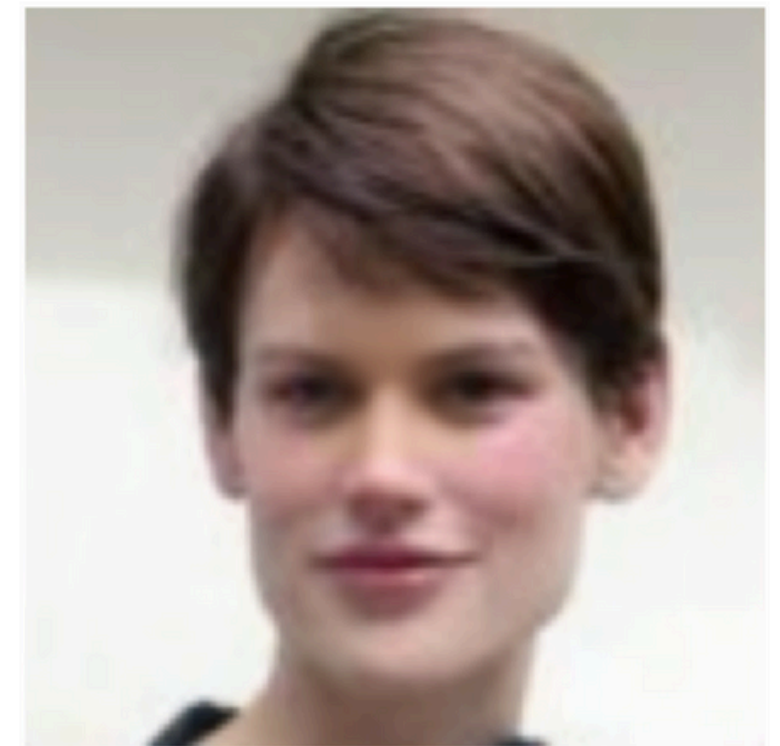
Original



Masked Input



Reconstructed



04

Future Improvement

Future Improvement

Using smaller patch_size, or using pixels as nodes in Graph.

Applying GAN on the top of CNN + GNN structure

Q&A